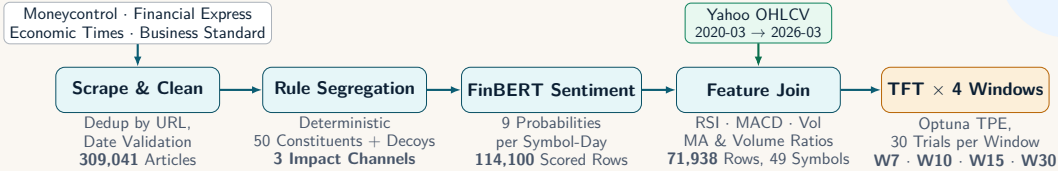
MarketFusion: A Multimodal Stock Forecaster

Indian Market News + OHLCV → A Leakage-Free, Reproducible Temporal Fusion Transformer



The Project in One Line

Six years of Indian market news are scraped, split by a **deterministic rule engine** into news that hits a stock *directly*, through its *sector*, or through the *global* tape, scored with FinBERT, fused with OHLCV technicals, and fed to a **Temporal Fusion Transformer** tuned per encoder length — the whole chain reproducible from a manifest.

Built and Trained On

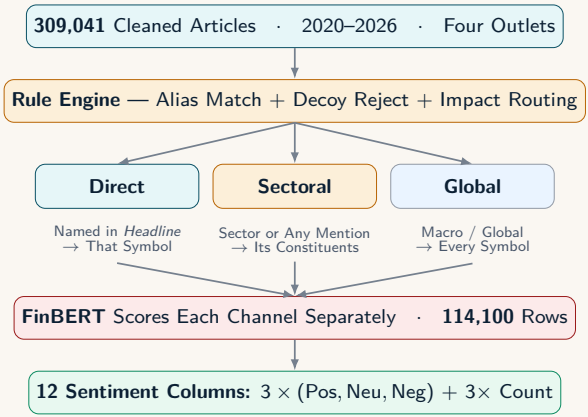
RTX 3050 — 4 GB VRAM

Bf16-Mixed

Grad Accum ×2 **Resumable Optuna**

Every stage checkpoints and resumes.

The Data Idea: One Article Can Reach a Stock Three Different Ways



What One Training Row Looks Like

- **29 Past-Observed Reals** — the return target, **16** technicals, **12** sentiment (means *and* article counts: 0.8 from one article $\neq$ 0.8 from forty).
- **6 Known-Future** — calendar only; tomorrow's news is not knowable.
- **1 Static** — the symbol, as an embedding.

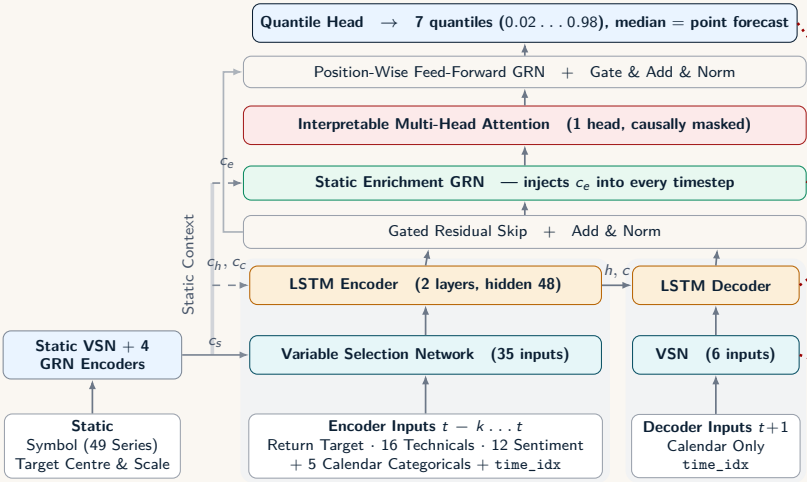
Leakage-Free by Construction

Train	Validation	Test
$\leq$ 2024-12-31 Fits Weights + Scalers	2025 H1 Early Stop	2025-07 $\rightarrow$ 2026-03 Metrics Only

Scalers are fit **per symbol on pre-2025 rows only**, and `adj_close_inr` is left unscaled — so a prediction rebuilds in rupees from the last close *before* the decoder starts.

The hard part is ambiguity. Names match longest-first against the 50 constituents *and* a decoy list, so Reliance Power, SBI Cards and ICICI Lombard never reach the constituent they contain. Ordinary-word aliases need capitalisation and a corporate cue; 26 self-test assertions cover every trap.

Inside the Model: Temporal Fusion Transformer, Block by Block



The Loss Is Also the Catch. Quantile loss gives free uncertainty bands — and on a near-signal-free target it drives the median toward a constant.

One Model, 49 Tickers. Series identity enters as a static embedding that conditions selection, LSTM state and enrichment — not 49 separate models.

Local vs. Long Range. The LSTM handles local order; attention reaches across the whole encoder. Encoder length is the one structural knob I swept: 7/10/15/30.

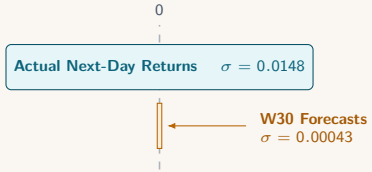
Interpretable by Design. The VSN emits softmax weights over all 35 inputs at every timestep, so I can read off *which* feature was used.

Selected W30 Config: **Hidden 48 · 1 Head · 2 LSTM Layers** | **Dropout 0.25 · LR 1.9e-4 · Clip 0.51** | **Batch 64×2 · Bf16-Mixed · Seed 72**

Results: A Controlled Benchmark — and Where It Points Next

Win.	MAPE %	MAE	Ratio	Dir. Acc
W7	**1.0494**	0.010480	1.004	0.5146
W10	1.0548	0.010525	1.008	**0.5178**
W15	1.0590	0.010567	1.012	0.4815
W30	1.0501	0.010475	1.003	0.4914

Held-out **2025-07** → **2026-03**. Ratio = model MAE ÷ random-walk MAE; **1.00** = **parity with the null**.



Forecasts Settle on the Conditional Median

What the Evaluation Establishes

- **Trustworthy Numbers.** Strict temporal split and pre-2025 scalers — 1.05% MAPE is real held-out error.
- **Level With the Null.** Ratios of 1.003–1.012 sit at the random walk, the correct benchmark here.
- **Well-Behaved Optimisation.** On a low-signal target the quantile optimum *is* the median — reached, not overfit.

Where This Points Next

- **Move the Target, Not the Model.** Weekly returns, volatility and direction labels carry far more signal.
- **Evaluate Where News Lands.** Score event and high-article-count days, not **71,938** mostly quiet rows.
- **Rank, Don't Level.** Ordering 49 symbols cross-sectionally is easier — and more tradeable — than levels.

Reading the table: all four windows land within 0.01 pp and their validation losses within 0.14% — neither encoder length nor tuning is the lever.
Beware F1: W30's best F1 (0.639) comes with *below*-chance direction (0.4914), purely from predicting "up" on 92% of bars.